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RESEARCH ARTICLE

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Abstract

The ICH E17 guideline describes regional pooling in multi-regional clinical trials (MRCTs) in terms of the similarity of effect modifier distributions, but provides no quantitative methodology. We propose the normalized Area Between Cumulative Distributions (nABCD), a planning-stage dissimilarity index defined as the Wasserstein-1 distance divided by twice the pooled interquartile range. Unlike the standardized mean difference (SMD), nABCD captures differences in scale and skewness in addition to location—distributional features that may drive treatment effect heterogeneity through non-linear effect modification. Because nABCD requires only baseline effect modifier distributions, it can be computed from any representative data source, including prior trials, disease registries, and real-world evidence (RWE) databases—enabling a common planning scenario in which a small-sample region identifies suitable pooling partners before the trial begins. We develop a dual-pathway clinical calibration: when prior evidence on conditional average treatment effect (CATE) sensitivity (L) is available, nABCD is translated into a maximum potential regional treatment effect difference ($\Delta_{\max}$) with bootstrap confidence intervals; when L is unknown, the framework reverse-calculates the required CATE sensitivity (L^*) at pre-specified clinical margins. Simulation studies across seven systematic scenarios demonstrated satisfactory bias and coverage of the percentile bootstrap estimator for non-negligible distributional differences at moderate sample sizes per region. We illustrate the framework through a hypothetical thrombolytic MRCT grounded in publicly available GUSTO-I individual patient data, with one region designated as a small-sample anchor and the remaining regions evaluated as pooling partners on two candidate effect modifiers (age and systolic blood pressure). Partner rankings differed markedly between the two effect modifiers, with several partners ranked similar on one but dissimilar on the other, demonstrating that all candidate effect modifiers must be evaluated jointly. A small subset of partners emerged as leading pooling candidates because they ranked low on both candidate effect modifiers and required clinically implausible CATE sensitivity to produce a meaningful treatment effect difference. The principal contribution of this work is to transform the previously qualitative judgment of "similar enough" into a reproducible quantitative basis for sponsor judgment, filling the methodological gap in ICH E17 implementation and supporting evidence-based regional pooling decisions from any representative population data source.

KEYWORDS
multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

1 INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become the standard paradigm for global pharmaceutical development.[1,2] This approach offers substantial benefits: accelerated timelines, broader generalizability, and earlier access to new therapies for patients worldwide. The International Council for Harmonisation (ICH) E17 guideline, adopted in 2017, established principles for planning and designing MRCTs, with a central assumption that treatment effects are generalizable across the target population.[3]

Several regulatory authorities expect demonstration of treatment effect consistency in regional subpopulations,[4,5] yet individual regional sample sizes in global MRCTs are often insufficient for such demonstration. A key strategy for addressing this challenge is the *regional pooling approach*, wherein regions with similar patient characteristics are grouped for analysis to enhance the ability to assess regional consistency.[3] The ICH E17 guideline describes pooling decisions in terms of the similarity of effect modifier distributions.[3]

An effect modifier is a baseline patient characteristic—such as age, disease severity, or genetic marker—for which the treatment benefit differs across subgroups. For example, if younger patients respond better to treatment than older patients, age is an effect modifier. When such heterogeneity exists, even if the drug works identically at the individual level, regions with different patient compositions may observe different average treatment effects. A region with predominantly younger patients would show larger benefits than a region with predominantly older patients, not because the drug works differently, but because the patient mix differs. This fundamental relationship underscores why effect modifier distributional similarity is critical to the validity of regional pooling.

Despite the regulatory importance of effect modifier distributional similarity, current practice lacks a standardized quantitative methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are "similar enough." Recent regulatory guidance has highlighted this gap. Song et al., writing from the China NMPA perspective on ICH E17 implementation, note the challenge of operationalizing pooling criteria without quantitative tools.[5] Long et al. further discuss basic considerations for consistency evaluation under ICH E17.[6]

This absence of quantitative tools is particularly acute at the *planning stage*, when sponsors must identify suitable pooling partners for small-sample regions. The pooling rationale requires quantitative evidence that partner regions' patient populations are "similar enough," yet no established method exists to provide such evidence. A workshop report by Matsushima et al. documented that regional imbalance in CRP-positive/MRI-negative status contributed to apparent inconsistency in the secukinumab MRCT.[4] This example illustrates the practical consequences when effect modifier distributional differences are not assessed quantitatively at the planning stage.

Although general-purpose comparison tools such as the standardized mean difference are routinely applied to baseline covariates,[7] they were not designed for assessing effect modifier distributional similarity across regions. As a single summary measure, the standardized mean difference cannot detect differences in variance or distributional shape. When effect modification is non-linear, these distributional features directly influence regional average treatment effects, and a comparison based on means alone will miss them.

This paper addresses these gaps by proposing the *normalized Area Between Cumulative Distributions* (nABCD) as a quantitative index for assessing effect modifier distributional similarity across regions. nABCD is defined as the Wasserstein-1 distance between two cumulative distribution functions, normalized by a robust scale measure (the pooled interquartile range) to achieve scale-free interpretation. Unlike the standardized mean difference, nABCD captures differences in variance and shape simultaneously. We estimate nABCD with bootstrap confidence intervals and link it to clinical interpretation through a calibration framework. We focus on continuous effect modifiers, for which the Wasserstein distance provides a theoretically grounded measure of distributional dissimilarity.

The remainder of this paper is organized as follows. Section 2 presents the methodological framework, including the index definition, clinical calibration, and estimation procedure. Section 3 describes a simulation study. Section 4 demonstrates the framework using publicly available individual patient data. Section 5 discusses implications, limitations, and future directions.

Abbreviations: ABCD, area between cumulative distributions; AMI, acute myocardial infarction; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; EDF, empirical distribution function; EU, European Union; GUSTO, Global Utilization of Streptokinase and t-PA for Occluded coronary arteries; ICH, International Council for Harmonisation; IPD, individual patient data; IQR, interquartile range; KL, Kullback–Leibler; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; nABCD, normalized area between cumulative distributions; NMPA, National Medical Products Administration; RWE, real-world evidence; SBP, systolic blood pressure; SMD, standardized mean difference; US, United States.

2 | METHODS

An effect modifier is a factor for which the treatment effect varies across subgroups. Formally, let the conditional average treatment effect (CATE) be denoted $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, where X is the effect modifier. When this function is non-constant, the average treatment effect observed in region r depends on the distribution F_r of the effect modifier in that region:

$$\bar{\tau}_r = \int \tau(x)\, dF_r(x). \tag{1}$$

Assessing pooling suitability therefore requires a measure of distributional difference between regions.

2.1 | Existing Approaches and Their Limitations

The most widely used measure for comparing covariate distributions across groups is the standardized mean difference (SMD), defined as

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_2}{S_{\text{pooled}}}, \tag{2}$$

where $\bar{X}_r$ denotes the sample mean in region r and S_{pooled} is the pooled standard deviation.[7] Because treatment effects in clinical trials are typically summarized as mean differences, SMD provides a scale-free comparison directly relevant to the quantities that drive clinical decision-making. However, by construction SMD captures only differences in location. Two distributions with identical means but markedly different variances—for example, $N(50, 5^2)$ and $N(50, 15^2)$—yield SMD = 0 despite a threefold difference in spread.

Distributional distance measures address this limitation by comparing distributions beyond summary statistics. Empirical distribution function (EDF) tests such as the Kolmogorov–Smirnov (KS) statistic, $D_{\text{KS}} = \sup_x |F_1(x) - F_2(x)|$, and related measures (Cramér–von Mises, Anderson–Darling) quantify differences in the full shape of distributions by comparing CDFs directly. Information-theoretic divergences such as Kullback–Leibler (KL) divergence,

$$D_{\text{KL}}(P \| Q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx, \tag{3}$$

take a density-based approach, measuring how one probability distribution differs from another.

However, each of these approaches has limitations when applied to comparing effect modifier distributions across regions. SMD captures only location differences and is blind to variance and shape. EDF-based statistics such as the KS statistic capture full distributional differences but lack a clinically interpretable scale and have no theoretical connection to treatment effect heterogeneity. KL divergence is asymmetric—comparing region A to region B yields a different value than comparing B to A—can diverge to infinity when empirical distributions do not fully overlap, and requires density estimation that is impractical with small regional samples. Crucially, none of these measures offers a theoretical link that would translate a distributional distance into a bound on regional treatment effect differences. These gaps motivate a measure that (i) captures distributional features beyond location, (ii) provides a clinically interpretable scale, and (iii) offers such a theoretical link. The nABCD index, defined in the following subsection, is designed to satisfy all three requirements.

2.2 | The nABCD Dissimilarity Index

Among distributional distances that could satisfy these three requirements, we adopt the Wasserstein-1 distance (W_1) for its unique theoretical connection to treatment effect heterogeneity (Proposition 2 below). The W_1 distance (sometimes called the Earth Mover's Distance) between two cumulative distribution functions F and G is defined as:[8]

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)|\, dx. \tag{4}$$

Geometrically, this equals the total area between the two CDFs (Figure 1). Unlike the standardized mean difference, the Wasserstein distance responds to changes in variance, skewness, and other distributional features.[9]

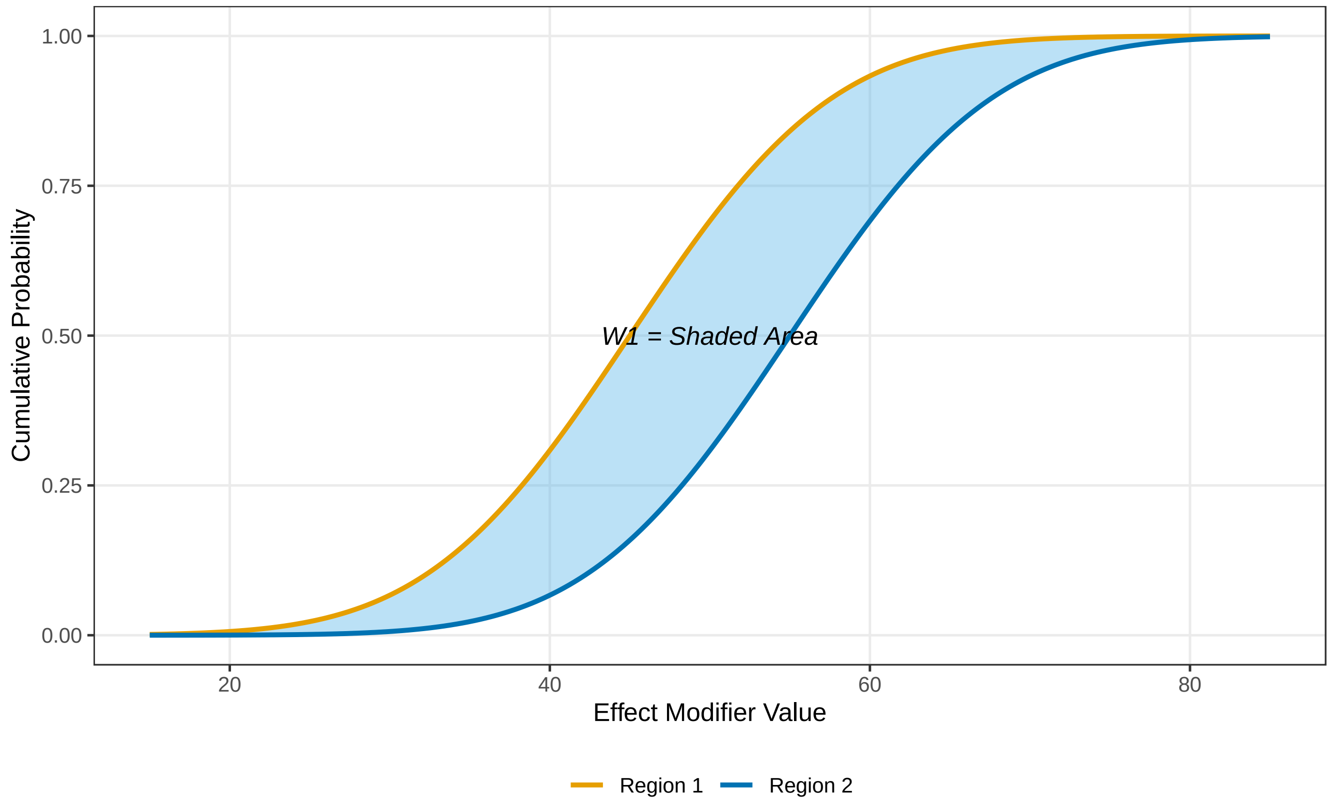

FIGURE 1 nABCD as the area between cumulative distribution functions. The shaded region represents the Wasserstein-1 distance $W_1(F_1, F_2)$, which equals the total area between the two CDFs. nABCD normalizes this area by twice the pooled IQR to achieve scale-free interpretation.

The *normalized Area Between Cumulative Distributions* (nABCD) is defined as the Wasserstein-1 distance normalized by twice the pooled interquartile range:

$$\text{nABCD}(F_1, F_2) = \frac{W_1(F_1, F_2)}{2 \cdot \text{IQR}_{\text{pooled}}}, \tag{5}$$

where $\text{IQR}_{\text{pooled}} = Q_{0.75}(F_{\text{pool}}) - Q_{0.25}(F_{\text{pool}})$ with $F_{\text{pool}} = \lambda F_1 + (1 - \lambda)F_2$ and $\lambda = n_1/(n_1 + n_2)$. The denominator normalizes W_1 by a scale estimator to achieve scale-free interpretation. We adopt the interquartile range (IQR) for its direct interpretability as the central 50% spread of the effect modifier distribution. The factor of 2 calibrates nABCD so that a one-IQR location shift yields nABCD = 0.5.

W_1 is a proper distance satisfying the triangle inequality.[8] Normalization by the pair-dependent pooled IQR does not preserve this property, but it enables comparison of distributional similarity across effect modifiers measured on different scales—a practical necessity when assessing multiple covariates simultaneously. nABCD thus serves as a dissimilarity index retaining non-negativity, symmetry, and identity of indiscernibles (Proposition 1).

Proposition 1. (Non-negativity). *For distributions with finite first moment, provided* $\text{IQR}_{\text{pooled}} > 0$ *(i.e., the pooled distribution is non-degenerate), nABCD* ≥ 0*, with equality if and only if* $F_1 = F_2$*.*

The specific choice of W_1 as the underlying distance is motivated by the Kantorovich–Rubinstein duality theorem,[8] which provides a theoretical bridge between distributional distance and treatment effect heterogeneity. The connection proceeds in three steps:

1. *W_1 as CDF area*: In one dimension, $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)| \, dx$ (equation 4), the total area between two CDFs. This area increases whenever the distributions differ in location, spread, or shape.

2. *Kantorovich–Rubinstein duality*: The W_1 distance has an equivalent dual representation:[8]

$$W_1(F_1, F_2) = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left| \int f\, dF_1 - \int f\, dF_2 \right|.$$

   That is, W_1 equals the largest possible difference in expected values of f between the two distributions, where the supremum is taken over all functions whose rate of change is bounded by 1 ($|f(x) - f(y)| \le |x - y|$ for all x, y, and such functions are called 1-Lipschitz).
3. *CATE application*: A function is Lipschitz continuous with constant L if its rate of change is bounded by L: $|\tau(x) - \tau(y)| \le L|x - y|$ for all x, y. In clinical terms, L controls how much the treatment effect can change per unit change in the effect modifier. Since τ/L is 1-Lipschitz, applying the duality yields:

$$|\bar{\tau}_1 - \bar{\tau}_2| \le L \cdot W_1(F_1, F_2). \tag{6}$$

This Lipschitz-dual characterization is specific to W_1. The W_2 distance, while admitting closed-form expressions for Gaussian families, does not possess it and therefore cannot provide such a bound. The alternative measures discussed in Section 2.1 similarly lack any analogous property.[10]

Combining the bound above with the nABCD definition (equation 5) yields the central result:

Proposition 2. (Connection to heterogeneity). *If the CATE function is Lipschitz continuous with constant L, then*

$$|\bar{\tau}_1 - \bar{\tau}_2| \le 2L \cdot IQR_{\text{pooled}} \cdot nABCD(F_1, F_2). \tag{7}$$

Proof. Substituting (5) into (6) yields $|\bar{\tau}_1 - \bar{\tau}_2| \le L \cdot W_1(F_1, F_2) = L \cdot 2 \cdot \mathrm{IQR}_{\text{pooled}} \cdot \mathrm{nABCD}(F_1, F_2) = 2L \cdot \mathrm{IQR}_{\text{pooled}} \cdot \mathrm{nABCD}(F_1, F_2)$. ☐

The Lipschitz assumption is appropriate when the CATE function varies smoothly with the effect modifier. For threshold or step-function CATEs, L can be arbitrarily large and the bound becomes conservative. In such cases the clinical calibration should be interpreted as a worst-case bound rather than a tight estimate (see Section 5).

2.3 | Estimation

Given samples $\{X_{1,i}\}_{i=1}^{n_1}$ from region 1 and $\{X_{2,j}\}_{j=1}^{n_2}$ from region 2, nABCD is estimated using empirical distribution functions. Since $\hat{F}_1$ and $\hat{F}_2$ are piecewise constant (right-continuous step functions), the integral in equation (4) reduces to a finite sum over the intervals defined by the combined order statistics:

$$\widehat{\mathrm{nABCD}} = \frac{\sum_{k=1}^{n_1+n_2-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)})}{2 \cdot \widehat{\mathrm{IQR}}_{\text{pooled}}}, \tag{8}$$

where $\hat{F}_r$ denotes the empirical CDF (right-continuous) of region r, and $x_{(1)} < \cdots < x_{(n_1+n_2)}$ are the combined order statistics. The asymptotic distribution of W_1 is non-standard. del Barrio et al.[11] showed that $\sqrt{n}\, W_1(\hat{F}_n, F) \xrightarrow{d} \int |B(F(x))|\, dx$, where B is a standard Brownian bridge. Because the quantiles of this limiting distribution depend on the unknown F, no universal critical values are available and direct construction of asymptotic confidence intervals is impractical. We therefore employ the nonparametric percentile bootstrap with $B = 2{,}000$ replicates.[12] In the two-sample setting, when $F_1 \ne F_2$ the Hadamard derivative of the L_1 functional is linear, so the ordinary nonparametric bootstrap is consistent.[13] Convergence occurs at rate $\sqrt{n_1 n_2/(n_1 + n_2)}$ (see Appendix A.2 for details). The estimator (8) is computed in $O((n_1 + n_2)\log(n_1 + n_2))$ time, dominated by sorting the combined order statistics.

2.4 | Interpretation and Clinical Calibration

A central feature of nABCD is its connection to treatment effect heterogeneity through Proposition 2. This connection provides the basis for a clinically grounded interpretation framework rather than reliance on fixed numerical cutoffs.

2.4.1 | Clinical Calibration via the Heterogeneity Bound

From equation (7), the maximum potential difference in regional average treatment effects attributable to effect modifier distributional differences is:

$$\Delta_{\max} = 2L \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot \mathrm{nABCD}(F_1, F_2), \tag{9}$$

where L is the Lipschitz constant of the CATE function $\tau(x)$ with respect to the effect modifier. L serves as a sensitivity parameter controlling the worst-case treatment effect heterogeneity. When CATE sensitivity L has been independently estimated—for example, from a prior Phase 2 interaction analysis or a meta-analysis reporting explicit per-unit subgroup-specific treatment effects on the outcome scale—equation (9) provides a direct clinical-scale translation of the observed distributional difference. In practice at the MRCT planning stage, a numerical L is often unavailable even when prior subgroup evidence exists. Published subgroup reports typically document direction and significance of interaction rather than a per-unit slope, and estimates are sensitive to model choice and the reporting scale. The L^* reverse-calculation introduced next (Section 2.4.2) is therefore the more commonly applicable planning-stage pathway, while equation (9) becomes applicable once L is available in confirmatory or post-Phase-2 settings.

The clinical calibration procedure is as follows:

1. Identify candidate effect modifiers and compute nABCD with bootstrap CIs for each effect modifier across region pairs.
2. For each effect modifier, estimate or bound L from prior knowledge (e.g., subgroup analyses showing treatment effect varies by $\Delta\tau$ per unit change in the effect modifier, giving $L \approx \Delta\tau/\Delta x$).
3. Compute $\Delta_{\max}$ using equation (9). This represents the worst-case treatment effect difference attributable to the distributional difference.
4. Derive the bootstrap CI for $\Delta_{\max}$ from the nABCD confidence interval: $[\Delta_{\max,L}, \Delta_{\max,U}] = 2L \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot [\mathrm{nABCD}_L, \mathrm{nABCD}_U]$. This expresses inferential uncertainty directly on the clinical scale.
5. Report $\Delta_{\max}$ and its CI alongside the clinical context—for example, the overall treatment effect size, the non-inferiority margin, or the minimal clinically important difference. This information enables trial designers and regulatory scientists to exercise clinical judgment about pooling, considering the totality of evidence rather than a binary accept/reject decision.

This estimation-centered approach deliberately avoids forcing pooling decisions into a binary hypothesis testing framework. The rationale is threefold. First, the ICH E17 guideline describes "similar enough" as inherently context-dependent, and reducing this judgment to a rejection threshold risks oversimplification. Second, the uncertainty in L means that $\Delta_{\max}$ itself is subject to sensitivity analysis (Section 4), and a binary test cannot naturally accommodate this layered uncertainty. Third, consistent with recent calls to move beyond dichotomous statistical decisions,[14] presenting $\Delta_{\max}$ with its CI provides richer information for regulatory deliberation than a p-value or a reject/fail-to-reject outcome.

When regulatory agencies require a formal decision rule, the estimation framework can be adapted by declaring pooling acceptable if the upper bound of the 95% CI for $\Delta_{\max}$ falls below a pre-specified clinical margin Δ_{clin}. However, we recommend this as a supplementary rather than primary use of the nABCD framework.

The clinical calibration procedure is demonstrated with real clinical trial data in Section 4, where two candidate effect modifiers with contrasting levels of CATE sensitivity evidence illustrate both full calibration and planning-stage assessment workflows.

2.4.2 | Sensitivity Analysis When L Is Unknown

When L cannot be estimated from prior data—the most common situation at the MRCT planning stage—a reverse sensitivity analysis asks *what value of L would be required for a given nABCD to produce a clinically meaningful* $\Delta_{\max}$. Rearranging equation (9):

$$L^* = \frac{\Delta_{\mathrm{clin}}}{2 \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot \mathrm{nABCD}}, \tag{10}$$

where Δ_{clin} is a pre-specified clinically important treatment effect difference (e.g., the overall treatment effect or the non-inferiority margin). If the required L^* exceeds the largest plausible CATE sensitivity for the effect modifier under consideration, the distributional difference is unlikely to produce meaningful treatment effect heterogeneity, regardless of the true L. This reverse calculation is demonstrated in Section 4.

2.4.3 | Interpreting nABCD Without a Universal Cutoff

nABCD values have no universal cutoff that separates "similar" from "dissimilar" distributions. The appropriate interpretation depends on clinical context, specifically on how distributional differences translate into potential treatment effect heterogeneity. Two pathways are available. When L is estimable from prior evidence, equation (9) provides a direct clinical-scale translation via $\Delta_{\max}$. When L is unknown—the typical MRCT planning situation—the L^* reverse calculation (equation 10) quantifies the CATE sensitivity that would be needed for a given nABCD to produce a clinically meaningful $\Delta_{\max}$, and the sponsor assesses whether that required L^* falls within a plausible range for the effect modifier and drug class.

In either pathway, we recommend ranking candidate pooling partners by nABCD and reporting, for each, the point estimate, the percentile bootstrap CI, and either $\Delta_{\max}$ (when L is known) or L^* at pre-specified Δ_{clin} values (when L is unknown). This ranking-plus-calibration presentation provides the quantitative inputs that a sponsor, in consultation with clinical and regulatory advisors, needs to exercise judgment about which partner regions are suitable for pooling. The framework deliberately does not prescribe a nABCD cutoff, because the clinical consequence of a given distributional distance depends on L—a parameter external to the distributional comparison. As a rough aid to intuition only, under Gaussian location shift with equal variances, nABCD = 0.05, 0.15, and 0.30 correspond approximately to SMD = 0.2, 0.5, and 1.0σ (simulation scenarios S2, S3, and S4 in Table 1, with true nABCD values 0.073, 0.180, and 0.328 respectively). This correspondence is provided for scale familiarity, not as a decision rule.

3 | SIMULATION STUDY

We conducted simulation studies to evaluate the estimation properties of the nABCD estimator: bias, variability, and coverage probability of bootstrap confidence intervals. These properties are essential for the clinical calibration framework, as reliable estimation of nABCD directly determines the quality of $\Delta_{\max}$ estimates. We assessed performance across a range of scenarios relevant to MRCT applications.

3.1 | Simulation Design

3.1.1 | Scenarios

We designed seven systematic scenarios for methodological validation, examining controlled distributional differences across location, scale, and combinations thereof. Table 1 summarizes these scenarios with their true nABCD values.

TABLE 1 Simulation scenarios with clinical motivations.

ID	Description	Distribution 1	Distribution 2	True nABCD
S1	Null (identical)	$N(50, 10^2)$	$N(50, 10^2)$	0.000
S2	Location 0.2σ	$N(50, 10^2)$	$N(52, 10^2)$	0.073
S3	Location 0.5σ	$N(50, 10^2)$	$N(55, 10^2)$	0.180
S4	Location 1.0σ	$N(50, 10^2)$	$N(60, 10^2)$	0.328
S5	Scale $1.5\times$	$N(50, 10^2)$	$N(50, 15^2)$	0.122
S6	Skew (Log-normal)	$N(50, 10^2)$	$\text{LogN}(\mu, \sigma)$	0.304
S7	Location + Scale	$N(50, 10^2)$	$N(55, 15^2)$	0.175

The Log-normal in S6 is parameterized to match mean 50 with CV $\approx$ 53%, typical of clinical biomarkers such as ALT. S7 combines location and scale differences, representing the most realistic MRCT pattern.

3.1.2 | Simulation Parameters

For each scenario, we generated samples of size n = 50, 100, and 200 per region, reflecting sample sizes typical in MRCT regional subgroups. We performed 10,000 replications per scenario–sample size combination to ensure stable estimates of operating characteristics, with Monte Carlo standard errors below 0.005 for all reported proportions. Bootstrap confidence intervals were computed using $B = 2{,}000$ resamples. All simulations were conducted in R version 4.5.1. True nABCD values for all scenarios were computed via Monte Carlo integration ($n_{\mathrm{MC}} = 10^6$) using the mixture IQR, avoiding closed-form approximations that may use component rather than pooled IQR.

3.1.3 | Evaluation Metrics

Consistent with the estimation-centered framework, we evaluated:

1. *Bias*: $\mathrm{Mean}(\widehat{\mathrm{nABCD}}) - \mathrm{nABCD}$
2. *Root mean squared error (RMSE)*: $\sqrt{\mathrm{Mean}[(\widehat{\mathrm{nABCD}} - \mathrm{nABCD})^2]}$
3. *Coverage probability*: Proportion of 95% bootstrap CIs containing the true value
4. *CI width*: Mean width of bootstrap CIs, reflecting estimation precision

3.2 | Results

3.2.1 | Point Estimation and Coverage

Table 2 presents the bias of the nABCD estimator across scenarios and sample sizes. The estimator showed positive bias under the null hypothesis (S1), with bias decreasing from 0.093 at $n = 50$ to 0.047 at $n = 200$. This positive bias is attributable to the non-negative nature of the Wasserstein distance. Even when true nABCD equals zero, sampling variability produces positive estimates.

TABLE 2 Bias of nABCD estimator by scenario and sample size.

Scenario	True nABCD	n = 50	n = 100	n = 200
S1 (Null)	0.000	0.093	0.066	0.047
S2 (0.2σ)	0.073	0.039	0.020	0.009
S3 (0.5σ)	0.180	0.010	0.003	0.001
S4 (1.0σ)	0.328	0.006	0.003	0.001
S5 (Scale)	0.122	0.028	0.014	0.007
S6 (Skew)	0.304	0.015	0.008	0.003
S7 (Loc+Scale)	0.175	0.023	0.011	0.006

Bias is defined as $\mathrm{Mean}(\widehat{\mathrm{nABCD}}) - \mathrm{nABCD}$. Results based on 10,000 replications with $B = 2{,}000$ bootstrap resamples.

For scenarios with true nABCD above approximately 0.1, bias was less than 0.02 in absolute value at $n \geq 100$, indicating satisfactory point estimation performance at practical sample sizes (Figure 2). For S4 (1.0σ location shift), bias was negligible (0.006 at $n = 50$, 0.001 at $n = 200$), confirming accurate estimation even for large distributional differences. For scenarios near the boundary (S1, S2), positive bias was more pronounced and diminished slowly with sample size—a well-known phenomenon for non-negative statistics where sampling variability inflates estimates when the true value is close to zero.

Table 3 presents coverage probabilities of the 95% bootstrap confidence intervals. For scenarios with true nABCD above approximately 0.1, coverage was near-nominal at $n \geq 100$: S3 (0.947–0.951), S4 (0.950–0.957), S6 (0.953–0.954), and S7 (0.935–0.944). S4 (1.0σ location shift) showed particularly stable coverage across all sample sizes, with no degradation at larger n, confirming satisfactory bootstrap performance even for large distributional differences. S5 (scale) showed moderate undercoverage at $n = 50$ (0.856) that improved to near-nominal at $n = 200$ (0.933). S2 (small location shift) showed undercoverage at $n = 50$ (0.652) due to its near-boundary true value, improving to 0.941 at $n = 200$.

Coverage for S1 (null) is not reported because its true nABCD (0.000) lies at the parameter space boundary, where the non-negativity constraint on nABCD produces persistent positive bias that prevents the confidence interval from containing the true value. This does not indicate estimator failure—the estimator is consistent—but rather that sample sizes well beyond 200 per region would be needed for reliable inference when the true distributional difference is negligible.

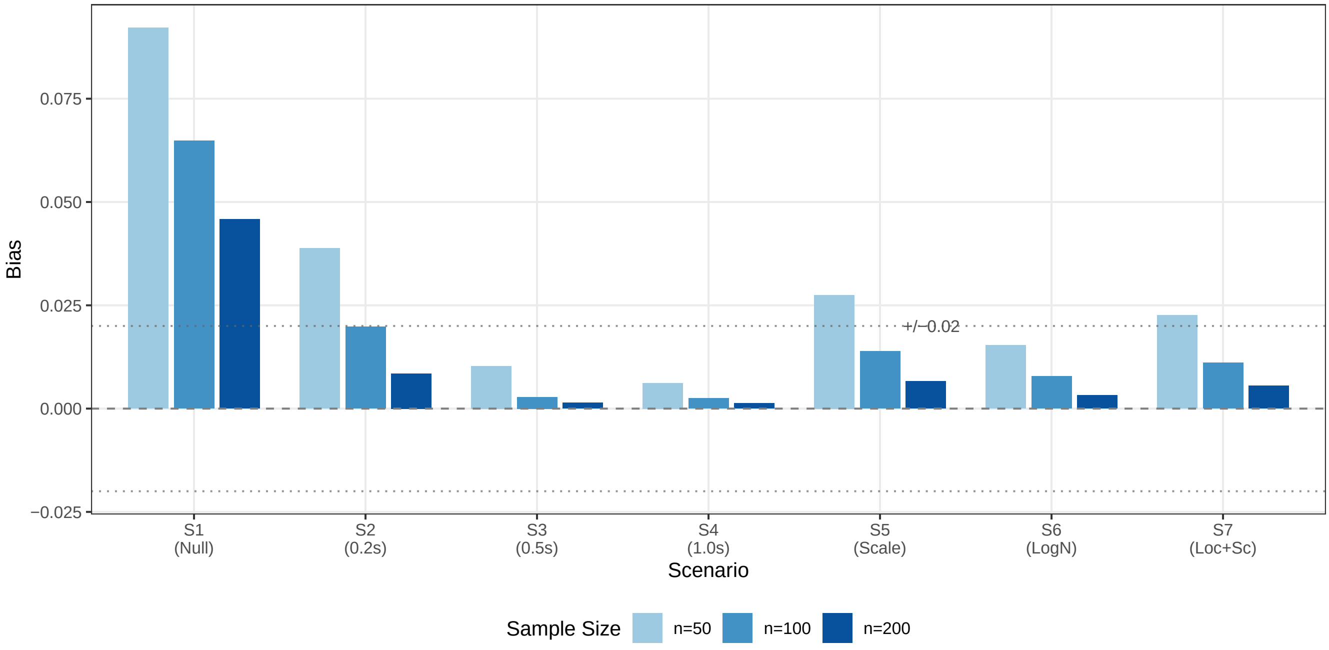

FIGURE 2 Bias of nABCD estimator by scenario and sample size. Horizontal dashed lines indicate ± 0.02 bias threshold. For scenarios with true nABCD above 0.1, bias is less than 0.02 at $n \geq 100$. Near-boundary scenarios (S1, S2) show larger positive bias due to the non-negativity constraint.

We also evaluated BCa (bias-corrected and accelerated) confidence intervals in preliminary experiments. BCa intervals showed consistently lower coverage than percentile intervals across all scenarios, particularly for near-boundary scenarios. The BCa correction overcorrects for nABCD because the statistic is bounded below by zero, causing the acceleration parameter to distort the quantile adjustment. Based on these findings, we adopt percentile bootstrap as the primary inference method.

Monte Carlo standard errors for all reported coverage probabilities were below 0.005 at 10,000 replications, ensuring that observed differences across scenarios and sample sizes reflect genuine operating characteristic differences rather than simulation noise.

TABLE 3 Coverage probability of 95% percentile bootstrap confidence intervals.

Scenario	$n = 50$	$n = 100$	$n = 200$
S2 (0.2σ)	0.652	0.881	0.941
S3 (0.5σ)	0.947	0.951	0.947
S4 (1.0σ)	0.950	0.950	0.957
S5 (Scale)	0.856	0.910	0.933
S6 (Skew)	0.953	0.954	0.954
S7 (Loc+Scale)	0.917	0.935	0.944

Coverage under S1 (Null) is not reported because the true nABCD lies at the boundary of the parameter space (0), where positive bias from the non-negativity constraint prevents meaningful coverage assessment.
For S4, coverage was near-nominal across all sample sizes (0.950–0.957), confirming satisfactory performance even for large distributional differences. The undercoverage for S2 at small sample sizes (0.652 at $n = 50$) and S5 (0.856 at $n = 50$) reflects boundary proximity effects that diminish with increasing n.

3.2.2 | Estimation Precision

Figure 3 summarizes the estimation quality across scenarios and sample sizes, presenting both coverage probabilities and mean CI widths.

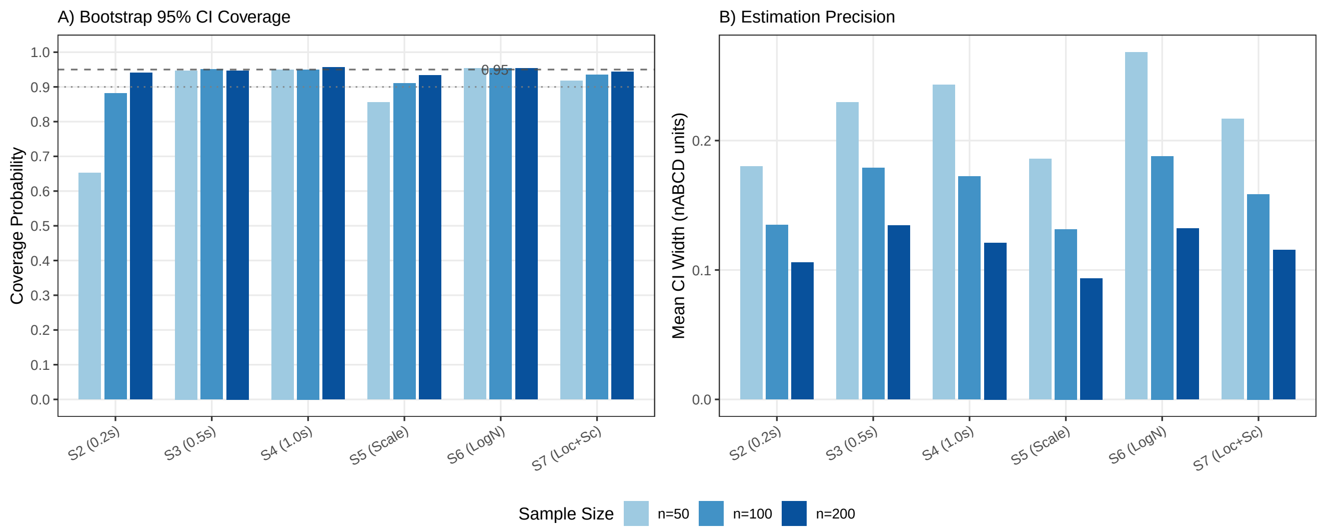

FIGURE 3 Estimation quality of nABCD bootstrap inference. Panel (A) shows coverage probability of 95% bootstrap confidence intervals, with dashed and dotted lines indicating the nominal 0.95 and minimum acceptable 0.90 levels. Panel (B) shows mean CI width in nABCD units. Coverage exceeds 0.90 for most scenarios at $n \geq 100$, and CI width narrows consistently with increasing n.

Table 4 presents the RMSE and mean CI width across scenarios. RMSE decreased consistently with increasing n, reaching values below 0.05 for all scenarios at $n = 200$. Mean CI width narrowed proportionally, with CIs at $n = 100$ typically spanning 0.11–0.19 in nABCD units. In the clinical calibration framework, this CI width propagates to $\Delta_{\max}$. For example, with $L = 0.01$/yr and IQR = 17 (values representative of age in the GUSTO-I application), a CI width of 0.13 in nABCD corresponds to a $\Delta_{\max}$ CI width of $2 \times 0.01 \times 17 \times 0.13 = 0.044$ (4.4%pt)—a level of precision sufficient for meaningful clinical calibration.

TABLE 4 RMSE and mean CI width by scenario and sample size.

	RMSE			Mean CI Width		
Scenario	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
S1 (Null)	0.099	0.069	0.049	0.16	0.11	0.08
S2 (0.2σ)	0.061	0.044	0.032	0.18	0.13	0.11
S3 (0.5σ)	0.067	0.048	0.035	0.23	0.18	0.13
S4 (1.0σ)	0.062	0.043	0.030	0.24	0.17	0.12
S5 (Scale)	0.053	0.036	0.025	0.19	0.13	0.09
S6 (Skew)	0.070	0.049	0.034	0.27	0.19	0.13
S7 (Loc+Scale)	0.062	0.043	0.030	0.22	0.16	0.12

CI width is defined as mean of (upper − lower) across 10,000 replications.

Under the null hypothesis (S1), the positive bias produces CIs that are shifted upward from zero. At $n = 50$, the mean estimate was 0.093 with mean CI width 0.16, and at $n = 200$, the mean estimate decreased to 0.047 with mean CI width 0.08. This bias is relevant for clinical calibration. When the true distributional difference is zero, the estimator will suggest a small positive $\Delta_{\max}$. Practitioners should be aware of this conservative tendency, particularly at smaller sample sizes.

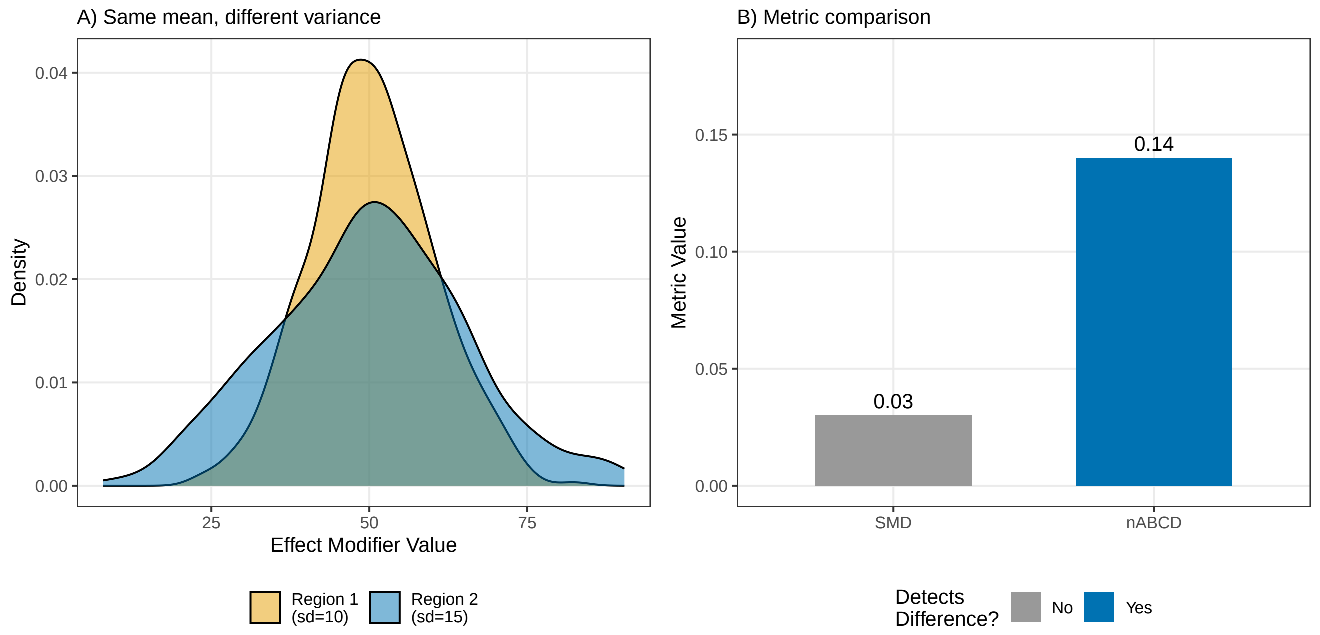

FIGURE 4 Comparison of nABCD and SMD across distributional difference types. nABCD responds to scale (S5) and skew (S6) differences where SMD remains near zero, demonstrating its advantage for full distributional comparison.

3.2.3 | Comparison with Standardized Mean Difference

Table 5 compares the sensitivity of nABCD and SMD to different types of distributional differences. While both metrics respond to location shifts, SMD is insensitive to differences in variance and shape—the very types of distributional differences that may drive treatment effect heterogeneity through non-linear CATE functions.

TABLE 5 Sensitivity comparison: nABCD versus SMD at $n = 100$.

Scenario	nABCD (mean $\pm$ SD)	SMD (mean $\pm$ SD)	Implication
S3 (Location)	0.183 ± 0.048	0.50 ± 0.14	Both detect
S5 (Scale only)	0.136 ± 0.033	0.00 ± 0.14	Only nABCD detects
S6 (Skew only)	0.312 ± 0.048	0.00 ± 0.14	Only nABCD detects

SMD depends only on the difference in means, making it insensitive to differences in variance and skewness. nABCD captures the full distributional difference. S6 is particularly informative because a large distributional difference (nABCD = 0.31) is entirely invisible to SMD.

In summary, the simulation study across seven scenarios demonstrates that the nABCD estimator provides reliable estimation with the following properties at sample sizes typical in MRCT regional subgroups:

1. *Bias*: Less than 0.02 for most non-null scenarios at $n \geq 100$, with positive bias largest near the boundary (S1: 0.066 at $n = 100$) and negligible for scenarios with true nABCD above 0.1 (S3: 0.003, S4: 0.003 at $n = 100$)
2. *Coverage*: 0.88–0.96 at $n \geq 100$ for scenarios with true nABCD ≥ 0.1 (S3–S7), and S6 (skew) and S7 (location + scale) achieved coverage between 0.93–0.95
3. *Precision*: RMSE below 0.06 and CI width below 0.19 at $n \geq 100$
4. *Sensitivity*: Captures location, scale, and skewness differences where SMD is limited to location only

Scenario S3 (0.5σ location shift, true nABCD = 0.180) exemplifies the operating characteristics at clinically relevant effect sizes. Bias was negligible (+0.003 at $n = 100$), coverage was nominal (0.951), and CI width (0.18) translated to a Δ_{max} CI width of $2 \times 0.01 \times 17 \times 0.18 = 0.061$ (6.1%pt) when calibrated with $L = 0.01$/yr and IQR = 17 (age-scale parameters representative of the GUSTO-I application)—a level of precision sufficient for informed regulatory deliberation.

These estimation properties ensure that nABCD, when combined with the clinical calibration framework, provides Δ_{max} estimates of sufficient quality for regulatory deliberation. We recommend $n \geq 100$ per region for reliable estimation and inference.

4 | APPLICATION: HYPOTHETICAL THROMBOLYTIC MRCT USING GUSTO-I DATA

We illustrate the nABCD framework through a hypothetical scenario grounded in publicly available data from GUSTO-I, a large international trial in acute myocardial infarction (AMI). A sponsor is developing a novel thrombolytic agent (Drug T) for AMI and is planning a Phase 3 MRCT across multiple regions. At the planning stage, the sponsor uses IPD from GUSTO-I ($N = 40{,}830$; 16 regions)[15] to assess distributional similarity of candidate effect modifiers across regions and to identify suitable pooling partners for a designated anchor region. GUSTO-I was not designed as an MRCT and its regions are anonymized. It serves as a convenient publicly available data source for methodological demonstration, and the anonymization precludes demographic or geographic interpretation of the distributional patterns. In practice, sponsors would assemble regional effect modifier distributions from the most representative and current sources available.

4.1 | Scenario Design and Effect Modifier Selection

For this illustrative exercise, the sponsor identifies two candidate effect modifiers for Drug T's treatment effect in AMI: age and systolic blood pressure (SBP). Drug T is a novel agent. Its Phase 2 programme has suggested age and SBP as candidate effect modifiers on the basis of exploratory analyses and biological plausibility, but the precision of these analyses is insufficient for numerical estimation of L. At this planning stage the sponsor therefore has no quantitative prior on the CATE sensitivity L for either candidate effect modifier, and class-level indirect evidence is of limited value as a numerical substitute. For **age**, the Fibrinolytic Therapy Trialists' (FTT) collaborative meta-analysis of existing fibrinolytic agents[16] concluded benefit is present *"irrespective of age"* and does not report a per-year CATE slope on the absolute mortality scale, so class transferability would not yield a defensible numerical prior for Drug T. For **SBP**, no class-level quantitative estimate of CATE sensitivity is available at all. We therefore treat L as unknown a priori for both candidate effect modifiers, and apply the L^* reverse-calculation pathway (equation 10) to both.

The sponsor selects Region 8 as the anchor region—representing, for instance, a regulatory market where local efficacy evidence is required but sample size is limited. Region 8 comprises $n = 2{,}916$ patients in GUSTO-I, with baseline characteristics summarized in Table 6. The exercise of identifying Region 8's suitable pooling partners simulates the decision a sponsor would face when seeking regional efficacy assessment through pooling as described in ICH E17.[3]

TABLE 6 Region 8 baseline characteristics (GUSTO-I).

Candidate effect modifier	Mean	SD	Skewness	IQR_{pooled}
Age (years)	60.2	12.1	−0.17	17–18
SBP (mmHg)	132.4	22.9	0.20	30–34

$n = 2{,}916$. IQR_{pooled} ranges reflect variation across the 15 partner-region pairs.

4.2 | Distributional Assessment: nABCD Across 15 Partner Regions

We computed nABCD between Region 8 and each of the remaining 15 regions for both candidate effect modifiers, with percentile bootstrap 95% confidence intervals ($B = 2{,}000$). Table 7 presents the results. Figure 5 displays these results as a forest plot with bootstrap CIs, with partners ordered by ascending age nABCD to aid visual comparison.

TABLE 7 nABCD values (with 95% percentile bootstrap CIs) for Region 8 versus each partner region (GUSTO-I), listed by region number.

Partner	n	$\mathrm{nABCD_{age}}$ [95% CI]	$\mathrm{nABCD_{SBP}}$ [95% CI]
R1	2188	0.018 [0.014, 0.028]	0.058 [0.044, 0.079]
R2	2952	0.061 [0.044, 0.079]	0.015 [0.012, 0.030]
R3	2030	0.076 [0.057, 0.095]	0.051 [0.035, 0.074]
R4	2876	0.016 [0.010, 0.032]	0.042 [0.029, 0.060]
R5	1909	0.011 [0.010, 0.026]	0.056 [0.038, 0.077]
R6	1585	0.026 [0.018, 0.040]	0.050 [0.032, 0.076]
R7	3150	0.011 [0.008, 0.026]	0.082 [0.064, 0.098]
R9	3123	0.017 [0.011, 0.032]	0.110 [0.091, 0.130]
R10	1717	0.038 [0.022, 0.058]	0.062 [0.045, 0.085]
R11	2491	0.035 [0.021, 0.052]	0.104 [0.082, 0.122]
R12	4352	0.024 [0.013, 0.040]	0.097 [0.083, 0.117]
R13	2297	0.034 [0.022, 0.051]	0.037 [0.023, 0.061]
R14	3437	0.020 [0.012, 0.036]	0.064 [0.052, 0.082]
R15	2576	0.017 [0.011, 0.034]	0.074 [0.057, 0.093]
R16	1231	0.050 [0.034, 0.071]	0.071 [0.051, 0.095]

Point estimates with 95% percentile bootstrap CIs (B = 2,000, seed = 2026). Regions are listed by region number, and ordering carries no implicit ranking. Bootstrap CIs quantify estimation uncertainty. Narrow CIs lend greater confidence to individual nABCD values, while overlapping CIs indicate partners whose relative position is not sharply resolved by the data.

The two candidate effect modifiers produce different patterns of distributional similarity. For age, partner-region nABCD point estimates span a narrow range, from 0.011 (R5, R7) at the low end up to 0.076 (R3) at the high end, and eleven of the fifteen partners have age nABCD below 0.040. For SBP, point estimates span a wider range, from 0.015 (R2) up to 0.110 (R9), with the majority of partners falling in the 0.050–0.110 interval. SBP distributions therefore exhibit greater between-region heterogeneity than age distributions across GUSTO-I, and the ranking of partners by each candidate effect modifier differs accordingly.

Several partners fall in the middle of each ranking where uncertainty intervals overlap substantially. R16's age nABCD (0.050) is the 13th-smallest value with 95% CI $[0.034, 0.071]$, placing it in a region where its relative position is not sharply determined by the data—the CI spans the range covered by several partners with smaller point estimates. R6's age nABCD (0.026) is the 9th-smallest value but its SBP nABCD of 0.050 sits near the lower end of the SBP-nABCD distribution with 95% CI $[0.032, 0.076]$, similarly spanning a broad region of the SBP ranking. These mid-rank cases illustrate the value of reporting bootstrap CIs alongside point estimates. The ranking itself is accompanied by an uncertainty interval that informs how confidently one partner can be placed above or below another.

An instructive contrast emerges between R2 and R9. R2 has the second-largest age nABCD (0.061) but the smallest SBP nABCD (0.015), while R9 has the 4th-smallest age nABCD (0.017) but the largest SBP nABCD (0.110). A ranking based on either candidate effect modifier alone would reach opposite conclusions about these two regions. This asymmetry underscores the necessity of evaluating all candidate effect modifiers jointly, as visualized in Figure 6.

4.3 | Clinical Interpretation and Pooling Candidates

For both candidate effect modifiers, L is not quantified at the planning stage, so we apply the L^* reverse-calculation (equation 10) to determine the CATE sensitivity required for the observed distributional differences to produce clinically meaningful heterogeneity (Δ_{clin} = 1%pt and 2%pt on the absolute 30-day mortality scale). Table 8 presents the full results for all 15 partner regions, and the joint distributional pattern is visualized in Figure 6.

The joint assessment identifies three regions—R4, R6, and R13—that rank low on both candidate effect modifiers, with all six nABCD values lying in the lower portions of the observed ranges (age: 0.011–0.076; SBP: 0.015–0.110) (Table 8; Figure 6, lower-left quadrant). These three regions exhibit low nABCD on both modifiers, and their required L^* values (Table 8; Figure 7) lie near the lower end of the range that could reasonably be considered clinically plausible for thrombolysis in AMI based on the available class evidence. Accordingly, *the sponsor may reasonably prioritize R4, R6, and R13 as the leading candidates for pooling with Region 8*.

The framework does not assert binary "poolable" or "not poolable" designations. Rather, it supplies quantitative inputs—nABCD values, bootstrap confidence intervals (Table 7), and L^* sensitivities (Table 8)—that support the sponsor's judgment in consultation with clinical and regulatory advisors. Sponsors considering partners outside these leading candidates (R4, R6, R13)

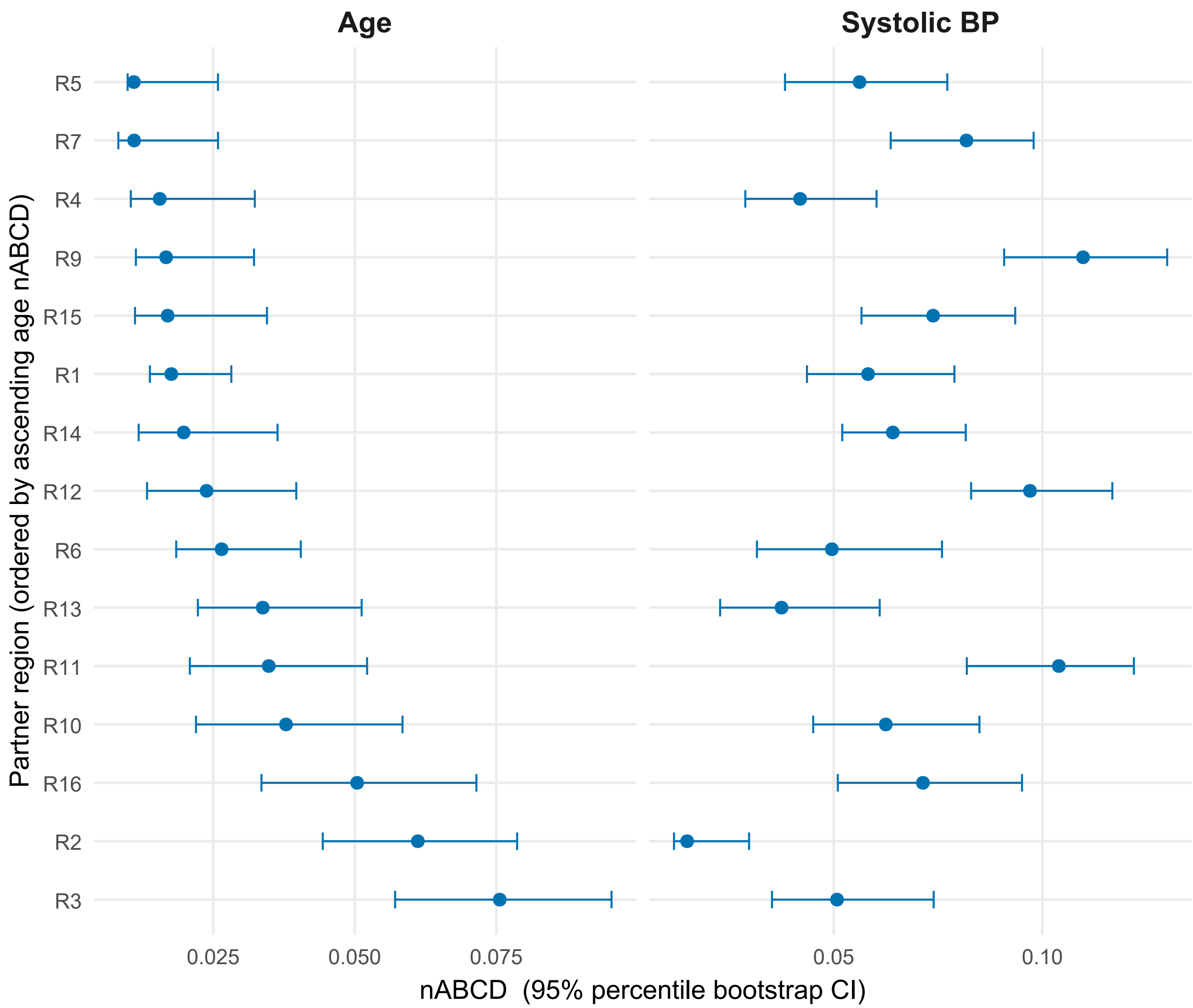

FIGURE 5 Forest plot of nABCD point estimates with 95% percentile bootstrap CIs for Region 8 versus each partner region, for age (left) and systolic blood pressure (right). Partners are ordered by ascending age nABCD to aid visual comparison. See Table 7 for values listed by region number. The bootstrap CI widths indicate the precision of each partner's estimated nABCD.

may examine the partner-specific L^* values and bootstrap CI widths to evaluate the precision and plausibility of pooling on each candidate effect modifier. We note that the GUSTO-I data were collected during 1990–1993, and demographic and clinical practice changes since then may limit the contemporary relevance of these specific region-level distributions; the application here serves strictly as a methodological illustration using publicly available IPD, not as an endorsement of GUSTO-I-era distributions as current references for thrombolytic-trial planning.

5 DISCUSSION

This paper addresses three methodological gaps identified in Section 2.1, demonstrating that nABCD: (i) captures scale and skewness differences in regional effect modifier distributions that are invisible to standardized mean differences, as confirmed by simulation scenarios with location-preserving variance and shape contrasts (S5, S6); (ii) translates distributional distance into a clinically interpretable scale via the heterogeneity bound $\Delta_{\max}$ (equation 9), providing a theoretical link to regional treatment

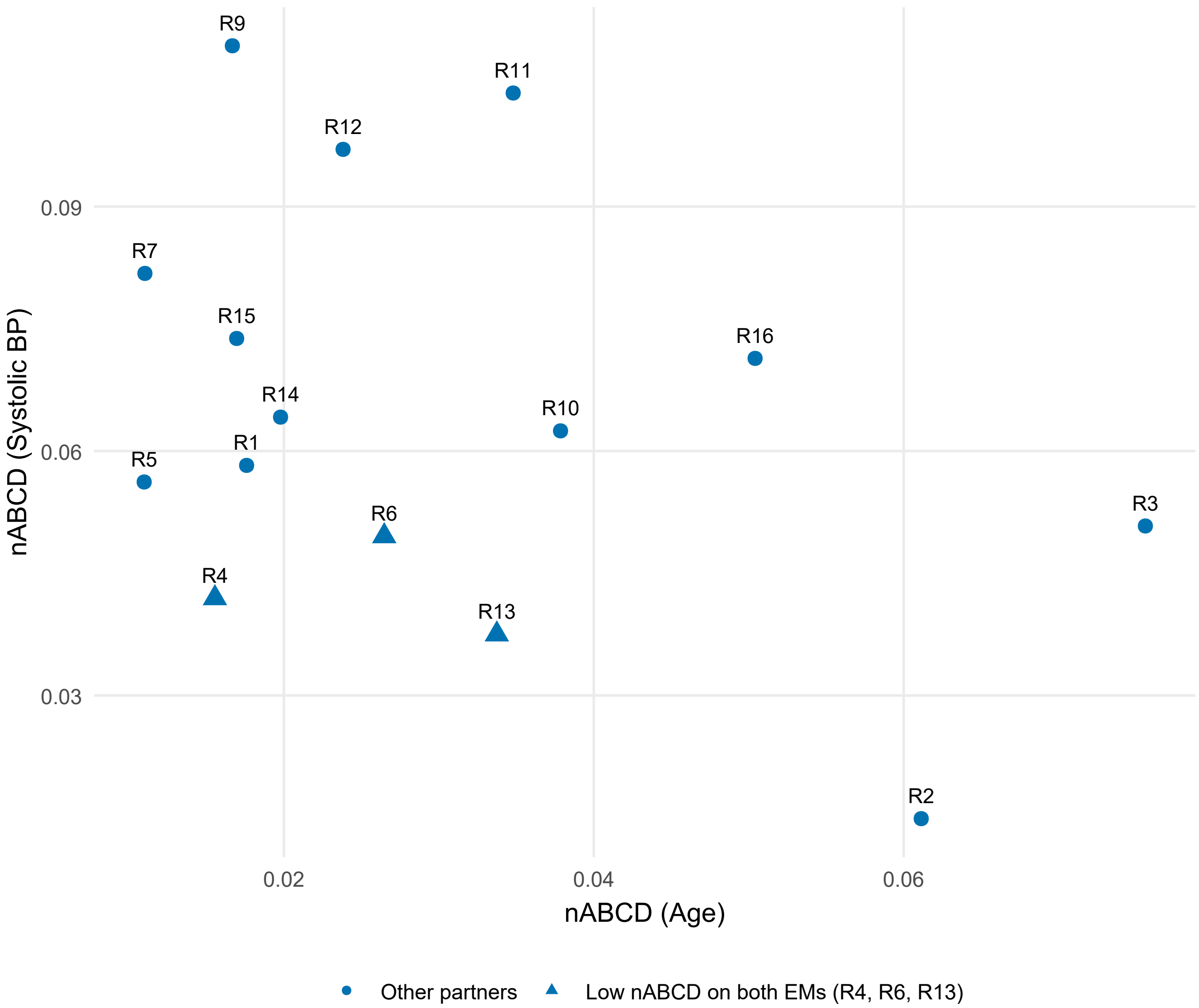

FIGURE 6 Joint distributional assessment for Region 8. Each point represents a partner region, positioned by nABCD for age (horizontal axis) and SBP (vertical axis). The plot presents distributional similarity as a continuum: partners closer to the lower-left corner rank low on both candidate effect modifiers (R4, R6, R13), partners further along one axis rank low on one but high on the other (R2, R9), and partners further on both axes rank high on both (R3).

effect heterogeneity that is absent in empirical-distribution-function tests such as the Kolmogorov–Smirnov statistic; and (iii) admits reliable percentile bootstrap inference for non-negligible distributional differences at moderate sample sizes ($n \geq 100$ per region, coverage 0.88–0.96 across S2–S7), while exhibiting characterized boundary behavior near the null that delimits the operational range of inference.

These methodological findings carry two interpretive implications for pooling decisions, illustrated by the GUSTO-I application as a worked example of the framework's operation rather than a substantive epidemiological claim. First, when multiple candidate effect modifiers are under consideration, all candidates must be evaluated jointly. A partner ranked similar on one effect modifier may not be similar on another, as exemplified by the contrast between R2 and R9. Restricting evaluation to a subset of candidates risks selecting partners whose distributional differences on the omitted modifiers would later compromise consistency. Second, partner selection cannot rest on nABCD ranking alone. The required CATE sensitivity L^* that would translate a given nABCD into a clinically meaningful $\Delta_{\max}$ varies markedly across partners and across candidate effect modifiers, so the same nABCD value carries different clinical weight depending on the partner-specific distributional differences. In the GUSTO-I example, R4,

TABLE 8 Clinical interpretation (L^* at Δ_{clin} = 1%pt) for all 15 partner regions, listed by region number. L^* is the minimum CATE sensitivity (per year for age, per mmHg for SBP) required for the observed distributional difference to produce a treatment effect difference of 1%pt on the absolute 30-day mortality scale.

Partner	$\text{nABCD}_{\text{age}}$	$\text{nABCD}_{\text{SBP}}$	L^* age (/yr)	L^* SBP (/mmHg)
R1	0.018	0.058	0.0155	0.0028
R2	0.061	0.015	0.0047	0.0105
R3	0.076	0.051	0.0038	0.0030
R4	0.016	0.042	0.0180	0.0038
R5	0.011	0.056	0.0256	0.0030
R6	0.026	0.050	0.0110	0.0030
R7	0.011	0.082	0.0247	0.0020
R9	0.017	0.110	0.0171	0.0015
R10	0.038	0.062	0.0072	0.0025
R11	0.035	0.104	0.0079	0.0016
R12	0.024	0.097	0.0116	0.0016
R13	0.034	0.037	0.0087	0.0042
R14	0.020	0.064	0.0142	0.0023
R15	0.017	0.074	0.0164	0.0021
R16	0.050	0.071	0.0057	0.0023

Regions are listed by region number, and ordering carries no implicit ranking. The L^* values quantify the CATE sensitivities at which these distributional differences would translate into clinically meaningful treatment effect heterogeneity.

R6, and R13 emerged as leading pooling candidates not solely because their nABCD values were low on both candidate effect modifiers, but also because the corresponding L^* values fell within ranges considered clinically plausible given the class-level evidence reviewed in Section 4. Pooling judgments therefore require both distributional ranking and clinical calibration, applied across the full set of candidate effect modifiers.

Section 2.1 identified three gaps in existing distributional measures. SMD captures only location. The KS statistic and related EDF tests lack a clinically interpretable scale and a theoretical link to treatment effect heterogeneity. KL divergence is asymmetric, can diverge under non-overlapping support, and requires density estimation impractical at small regional samples. nABCD addresses all three. Where SMD captures only the mean, our simulation confirmed that scale (S5) and skewness (S6) differences produced near-zero SMD yet non-negligible nABCD. Where the KS statistic offers no clinical-scale translation, the IQR normalization and the heterogeneity bound (equation 9) translate distributional distance into clinically meaningful units. Where KL divergence is asymmetric and prone to instability under limited sample overlap, nABCD is symmetric, finite for any pair of empirical distributions, and computable directly from CDFs without density estimation, while remaining amenable to bootstrap inference.

Two principal strengths follow from the dual-pathway clinical calibration. First, the framework adapts to where the trial sits on the spectrum of CATE sensitivity evidence, rather than imposing a fixed nABCD cutoff. At the MRCT planning stage, where L is typically unavailable for novel agents, the L^* reverse-calculation (equation 10) serves as the primary calibration tool. When L is available from prior evidence, the $\Delta_{\max}$ pathway (equation 9) provides a complementary calibration on the clinical scale. Second, nABCD's value as a distributional measure, capturing scale and skewness differences invisible to SMD, holds regardless of whether L is available. Clinical calibration enhances but does not replace this distributional assessment.

For practice, we recommend computing nABCD with bootstrap CIs at moderate sample sizes (Section 3) for every candidate effect modifier across region pairs, translating each into $\Delta_{\max}$ (when L is estimable) or L^* at pre-specified Δ_{clin} values (when L is unknown), and reporting these alongside clinically relevant benchmarks (overall treatment effect, non-inferiority margin) to support deliberation rather than binary decisions. When multiple candidate effect modifiers are evaluated, sponsors may adopt a conservative approach based on the maximum $\Delta_{\max}$ (or smallest L^*) across all effect modifiers and region pairs, or a totality-of-evidence approach in which the full collection—together with bootstrap uncertainty and the reliability of each L estimate—informs holistic judgment. The choice depends on regulatory context.

For policy, the framework's compatibility with diverse data sources is a distinctive practical advantage. Because nABCD requires only baseline effect modifier distributions, regional data can be assembled from prior trials, registries, electronic health records, or other real-world evidence (RWE) sources—an increasingly relevant capability as regulatory agencies promote RWE use in drug development (ICH E6(R3)). The framework is particularly relevant for regulatory submissions requiring regional subpopulation consistency.[4,5] Matsushima et al.'s[4] PMDA workshop documented how regional imbalance in CRP-positive/MRI-negative status caused apparent inconsistency in the secukinumab MRCT, underscoring the practical consequences when effect modifier distributional differences are not assessed prospectively.

These caveats notwithstanding, nABCD is one input within a holistic regulatory evaluation; biological plausibility, clinical relevance, and benefit–risk balance—captured by frameworks such as ICH E17 and the PMDA 3-layer approach—extend beyond what any single distributional measure can quantify.[3,4] Several limitations should be acknowledged:

1. *Scope.* The current formulation applies to continuous, univariate effect modifiers only. Extensions to categorical effect modifiers and multivariate joint evaluation—through, for example, higher-dimensional optimal transport or weighted combinations across modifiers—remain topics for future investigation.
2. *Behavior near the null.* As a non-negative function of the Wasserstein distance, nABCD exhibits positive bias when the true value lies at or near the boundary of the parameter space.[17] In S1 (null; true nABCD = 0.000), bias was substantial (0.093 at $n = 50$, 0.047 at $n = 200$) and bootstrap coverage was zero at all sample sizes; we therefore do not report coverage for this boundary scenario and recommend interpreting nABCD estimates near zero with caution. In the interior of the parameter space (true nABCD $\gtrsim$ 0.1), percentile bootstrap inference is reliable, with coverage 0.88–0.96 at $n \geq 100$ across S2–S7 (Section 3.2). We accordingly recommend $n \geq 100$ per region for reliable point estimation and confidence intervals.
3. *Clinical calibration requires CATE sensitivity input.* The $\Delta_{\max}$ pathway requires an estimate of L, which may be unavailable for novel agents; the L^* pathway addresses this by reverse-calculating the required sensitivity at a pre-specified Δ_{clin}, but ranking-based pooling decisions then require sponsor judgment on what L^* values are clinically plausible. When L is carried forward from prior evidence (e.g., Phase 2 or class-level meta-analyses), the calibration assumes that CATE sensitivity transfers to the target setting, a clinical judgment that may benefit from sensitivity analysis over plausible L values. This paper provides the quantitative inputs but does not prescribe cutoffs for nABCD or for L^*.
4. *Conditioning caveat.* As with any covariate-based approach, nABCD assesses similarity only with respect to measured effect modifiers; unmeasured or unknown effect modifiers may contribute to treatment effect heterogeneity beyond what the index captures.

Beyond the scope and methodological extensions noted above, two further directions warrant investigation. The upstream identification of which baseline characteristics constitute relevant effect modifiers is itself a non-trivial problem deserving dedicated treatment. Bias-correction methods tailored to the boundary behavior of nABCD at small samples could extend the operational range of inference. The principal contribution of this paper is to supply a quantitative tool where one was previously absent. nABCD, combined with bootstrap confidence intervals and clinical calibration via $\Delta_{\max}$ or L^*, transforms the previously qualitative judgment of "similar enough" into a reproducible, quantitative basis for sponsor judgment, filling the methodological gap left by ICH E17 and enabling evidence-based, clinically grounded pooling decisions.

AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

R code for computing nABCD and reproducing the simulation study is available at [repository URL]. GUSTO-I individual patient data are available from the original investigators subject to applicable data sharing policies; the dataset is described in the original publication.[15]

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website. Supporting materials include: complete R code for nABCD computation, simulation scripts, and additional figures for realistic clinical scenarios.

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APPENDIX

A PROOFS AND MATHEMATICAL DETAILS

A.1 Proof of Proposition 1

Non-negativity follows directly from the non-negativity of the Wasserstein-1 distance: $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx \geq 0$. Since $\text{IQR}_{\text{pooled}} > 0$ for non-degenerate distributions, the ratio $\text{nABCD} = W_1/(2 \cdot \text{IQR}_{\text{pooled}}) \geq 0$.

For the identity of indiscernibles, we show both directions. If $F_1 = F_2$, then $|F_1(x) - F_2(x)| = 0$ for all x, so $W_1(F_1, F_2) = 0$ and hence nABCD = 0. Conversely, if nABCD = 0, then $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx = 0$. Since $|F_1(x) - F_2(x)| \geq 0$ and is right-continuous, $\int |F_1(x) - F_2(x)|\, dx = 0$ implies $F_1(x) = F_2(x)$ almost everywhere, and by right-continuity of CDFs, $F_1 = F_2$ everywhere.

A.2 Asymptotic Properties

Under standard regularity conditions, the empirical Wasserstein-1 distance $W_1(\hat{F}_1, \hat{F}_2)$ is a consistent estimator of $W_1(F_1, F_2)$.[11] Combined with the consistency of the empirical IQR, $\widehat{\text{nABCD}}$ is a consistent estimator of $\text{nABCD}(F_1, F_2)$.

In one dimension, the empirical W_1 distance converges at rate $O(n^{-1/2})$ without logarithmic correction.[11] For two-sample estimation with $F_1 \neq F_2$ and finite second moments, $\sqrt{n}(W_1(\hat{F}_{1,n_1}, \hat{F}_{2,n_2}) - W_1(F_1, F_2))$ converges in distribution to a mean-zero

Gaussian, where $n = n_1 + n_2$ with $n_1/n \to \lambda \in (0, 1)$. Since the empirical IQR also converges at rate $O(n^{-1/2})$ under standard density conditions, the delta method applied to the ratio $g(w, q) = w/(2q)$ yields asymptotic normality of $\widehat{\text{nABCD}}$:

$$\sqrt{n}\left(\widehat{\text{nABCD}} - \text{nABCD}(F_1, F_2)\right) \xrightarrow{d} N(0,\, \sigma^2_{\text{nABCD}}), \tag{A1}$$

where σ^2_{nABCD} depends on the asymptotic variances and covariance of the W_1 and IQR estimators. This result requires nABCD > 0 (i.e., $F_1 \neq F_2$). At the boundary $F_1 = F_2$, the parameter lies on the edge of the parameter space and standard asymptotics do not apply (see Section 5).

The bootstrap provides valid inference for the Wasserstein distance under mild conditions.[13] The percentile bootstrap confidence interval achieves asymptotically correct coverage for non-degenerate distributions.

B R CODE FOR NABCD COMPUTATION

Listing 1 R function for computing nABCD with bootstrap confidence intervals.

```r
compute_nABCD <- function(x1, x2) {
  pooled <- c(x1, x2)
  iqr_pooled <- IQR(pooled)
  if (iqr_pooled == 0) return(NA)
  all_vals <- sort(unique(pooled))
  F1 <- ecdf(x1); F2 <- ecdf(x2)
  w1 <- sum(diff(all_vals) *
    abs(F1(all_vals[-length(all_vals)]) -
        F2(all_vals[-length(all_vals)])))
  w1 / (2 * iqr_pooled)
}

nABCD_bootstrap <- function(x1, x2,
    B = 2000, conf = 0.95) {
  obs <- compute_nABCD(x1, x2)
  boot_vals <- replicate(B, {
    b1 <- sample(x1, replace = TRUE)
    b2 <- sample(x2, replace = TRUE)
    compute_nABCD(b1, b2)
  })
  alpha <- 1 - conf
  ci <- quantile(boot_vals,
    c(alpha/2, 1 - alpha/2), na.rm = TRUE)
  list(estimate = obs,
       ci_lower = ci[1], ci_upper = ci[2])
}
```

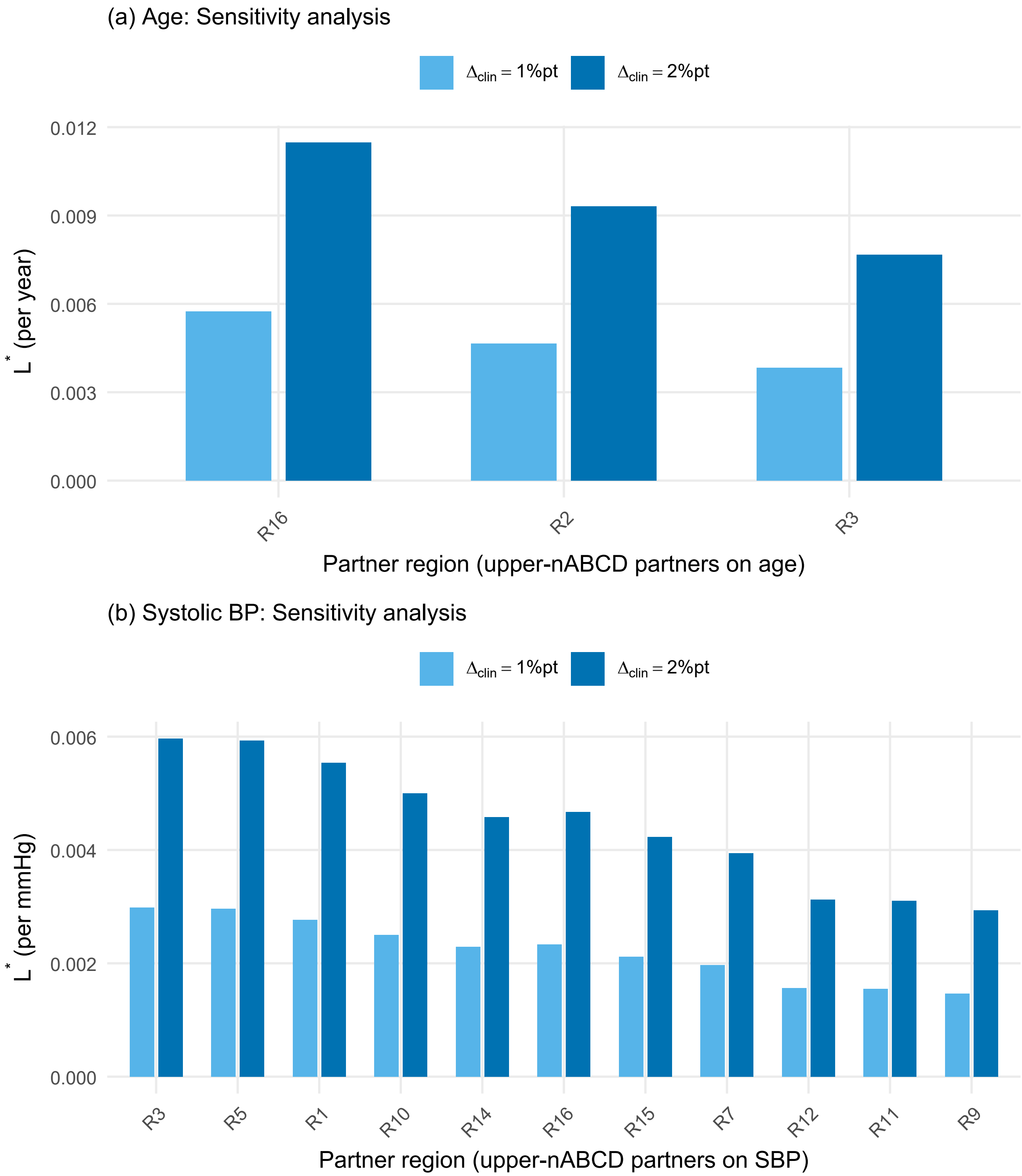

FIGURE 7 Clinical interpretation (L^*) for Region 8. Both panels show L^*—the minimum CATE sensitivity that would translate the observed distributional difference into a clinically meaningful treatment effect difference—at two clinical margin values (Δ_{clin} = 1%pt, 2%pt). (a) Age: L^* (per year) for the partners with the largest age nABCD (R16, R2, R3). (b) SBP: L^* (per mmHg) for the partners with the largest SBP nABCD values.